

POWER BI DOCUMENTATION & INSIGHTS

PRESENTATION

DASHBOARD LINK:

https://drive.google.com/file/d/145nnTYk_qykFgTX6XkjlFvXOsAT6W8qO/view?usp=sharing

SUMMARY:

The Medical Equipment Suppliers Dashboard is developed to analyze and visualize supplier information across different states, medical specialties, pharmacies, and business organizations. The project involves cleaning and preprocessing the raw dataset using Python to ensure data accuracy and consistency. The cleaned data is then imported into Power BI to create an interactive dashboard that presents meaningful insights through various visualizations.

The dashboard enables users to analyze supplier distribution, specialty-wise supplier counts, state-wise supplier analysis, business participation, and other key performance indicators (KPIs). Interactive charts, filters, and visual reports help users explore the data efficiently and support better decision-making. This project demonstrates the effective use of data analytics and business intelligence tools to transform raw healthcare supplier data into actionable insights.

DASHBOARD STRUCTURE : (Medical Equipment Suppliers Data Analysis)

Top Section (Key Performance Indicators)

This section provides a quick overview of the dataset through important summary metrics.

- **Total Medical Equipment Suppliers** – Displays the total number of suppliers available in the dataset.
- **Total Practice States** – Shows the number of states where suppliers are located.
- **Total Medical Specialties** – Indicates the total number of medical specialties covered by the suppliers.
- **Total Business Organizations** – Displays the total number of unique supplier business names.

Comparison & Trend Charts

This section helps users compare supplier information across different categories and identify trends.

- **Supplier Count by Medical Specialty** – Compares the number of suppliers available in each medical specialty.
- **State-wise Supplier Distribution** – Compares supplier counts across different states such as New York, California, Texas, and Florida.
- **Business-wise Supplier Participation** – Displays supplier participation for different business organizations.
- **Specialty-wise Supplier Comparison** – Helps compare supplier distribution among various medical specialties.

Relationship & Distribution Analysis

This section focuses on identifying relationships and understanding the distribution of supplier data.

- **Specialty-wise Supplier Distribution (Pie Chart)** – Shows the percentage distribution of suppliers across medical specialties.
- **Practice ZIP Code vs. Medical Specialty (Scatter Chart)** – Analysis the relationship between practice ZIP codes and medical specialties.
- **Telephone Information Analysis (Gauge/Donut Chart)** – Provides an overview of supplier contact information availability.
- **Supplier Details Table** – Displays detailed supplier information such as business name, practice city, specialty, state, and participation status for deeper analysis.

KEY INSIGHTS

- **High Supplier Concentration** – States such as California, New York, Texas, and Florida have the highest number of medical equipment suppliers.
- **Specialty-wise Analysis** – The dashboard identifies medical specialties with the highest supplier distribution.
- **Business Participation** – Different business organizations contribute varying numbers of suppliers to the dataset.
- **Geographical Distribution** – State and ZIP code analysis helps understand the regional distribution of suppliers.

- **Interactive Data Exploration** – Filters and slicers allow users to analyze data based on state, specialty, and business name.
- **Better Decision-Making** – The dashboard provides clear visual insights that support effective planning and informed business decisions.

BUSINESS RECOMMENDATIONS FOR DECISION MAKING

- **Focus on High-Demand States** – Increase supplier partnerships in states with the highest demand, such as California, New York, Texas, and Florida.
- **Expand Specialty Coverage** – Encourage supplier growth in medical specialties with lower supplier availability to improve healthcare access.
- **Strengthen Supplier Network** – Build relationships with reliable suppliers to ensure consistent availability of medical equipment.
- **Improve Geographic Coverage** – Expand supplier presence in underserved regions to achieve balanced distribution across all states.
- **Use Data for Strategic Planning** – Leverage dashboard insights to optimize procurement, resource allocation, and supplier management.
- **Monitor Supplier Performance** – Regularly track supplier participation and update the dashboard to support informed business decisions and continuous improvement.

BUSINESS INTERPRETATION:

- **Identifies supplier distribution** across different states and medical specialties.
- **Highlights high-performing regions** with a greater number of medical equipment suppliers.
- **Compares business participation** to understand supplier involvement across organizations.
- **Supports efficient resource planning** through interactive charts and data analysis.
- **Enables better decision-making** by providing clear and meaningful business insights.

FINAL STORY :

The Medical Equipment Suppliers Dashboard was developed to analyze and visualize medical supplier data through interactive reports and visualizations. The dashboard provides valuable insights into supplier distribution across different states, medical specialties, and business organizations. It enables users to identify trends, compare supplier performance, and explore data using interactive charts, KPIs, and filters. These insights support efficient resource planning, supplier management, and informed business decision-making. Overall, the dashboard transforms complex data into clear, meaningful, and actionable information for the healthcare sector.

